What does the evidence let us say?

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Evidence / ES: evidencia / PT: evidência

What was actually stated or measured. “An object is approaching.”

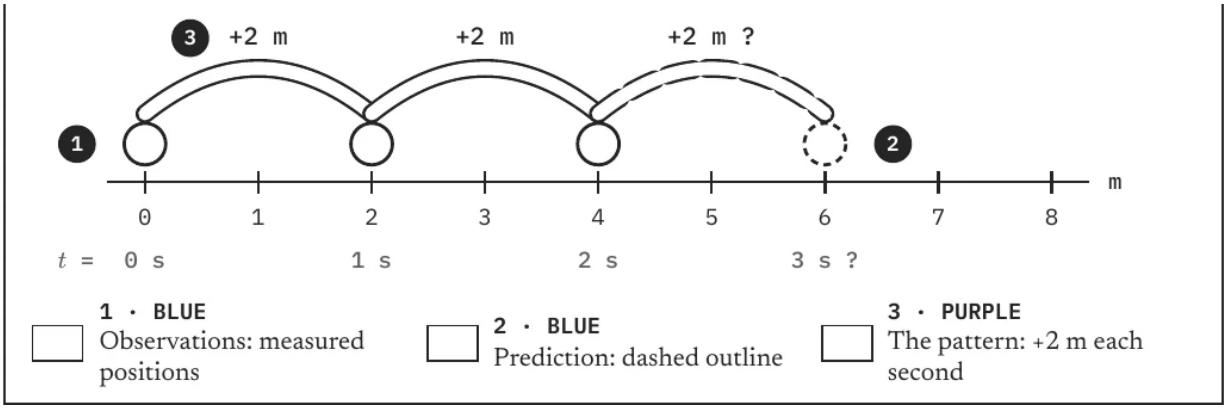
Also on your note page: Inference, Model, Testable question.

B READ THE RELATIONSHIP

New position = starting position + change per second × elapsed time

As you read, find: measured · predicted.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart is at 10 m, 8 m, and 6 m at 0, 1, and 2 s. Predict its position at 4 s, and name an observation that would challenge the model.

Given	x = 10, 8, 6 m t = 0, 1, 2 s
Rule	new x = start + change × time
Work	change = −2 m each second x (4 s) = 6 + (−2)(2)
Answer	2 m limit: a 3 m reading would challenge it

E SAVE FOR THE FINAL MEETING

The desk holds evidence (what was measured) and inference (what it suggests). A forecast for 2026-XJ must rest on the evidence: positions, times, and their uncertainty.

Read the physics: Evidence

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A telescope records where an object is at a known time. That is an observation. A future dot is a prediction, even if it follows a clear pattern.

02 HOW IT WORKS

Give each observed dot a time. Compare the gaps between dots. Equal gaps in equal times suggest constant velocity. Draw a dashed dot for the next position. It is still a guess based on a pattern.

03 WHAT TO CHECK

If a new dot misses the predicted position, check the measurement and the motion model.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at 10 m, 8 m, and 6 m at 0, 1, and 2 s. Predict its position at 4 s, and name an observation that would challenge the model. 2 m limit: a 3 m reading would challenge it.

Model check: New position = starting position + change per second × elapsed time.

05 CONNECT TO XJ

The desk holds evidence (what was measured) and inference (what it suggests). A forecast for 2026-XJ must rest on the evidence: positions, times, and their uncertainty.

SAY IT BEFORE YOU WRITE

Which dot did someone measure? The measured dot is ___. The predicted dot is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Evidence

ES: evidencia / PT: evidência

Meaning in my words	How can you tell?
How would you explain evidence to a friend? In my words, this means ___.	What reading supports the claim that the cart reached 4 m in 2 s? I can tell because ___.
Example or sketch	**Non-example and why**
How could two measurements support a claim that one cart accelerated more? My example shows ___.	"I expected our cart to win." Does that tell us which cart crossed first? I would change ___ because ___.

Inference

ES: inferencia / PT: inferência

Meaning in my words	How can you tell?
How would you explain inference to a friend? In my words, this means ___.	Wet tracks run from the door to a cart. Which part is your inference? I can tell because ___.
Example or sketch	**Non-example and why**
Two timed images show XJ closer to Earth. What can you reasonably infer? My example shows ___.	"The second image reads 4 km." Is that your inference or your reading? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Model

ES: modelo / PT: modelo

Meaning in my words	How can you tell?
How would you explain model to a friend? In my words, this means ___.	A cart moves 2 m each second. What are we assuming if we extend that pattern? I can tell because ___.
Example or sketch	**Non-example and why**
Where would your model put the cart one second after it passes 6 m going right? My example shows ___.	The cart hits a cushion. Would you still extend the same pattern? Why? I would change ___ because ___.

Testable question

ES: pregunta comprobable / PT: pergunta testável

Meaning in my words	How can you tell?
How would you explain testable question to a friend? In my words, this means ___.	What would you measure to answer “Does this cart speed up on the ramp?” I can tell because ___.
Example or sketch	**Non-example and why**
What question could two timed pictures of the same cart help you answer? My example shows ___.	“Which cart looks coolest?” What would you change so the class can test it? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a three-sighting timeline. Label the starting position, change per second, and one future position. Use a dotted mark for the forecast.

Labels you can use: measured · predicted.

Choose zero, direction, and scale.



Tell a partner

The measured dot is ___. The predicted dot is ___.

One more exposure

01

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Separate a sighting from a prediction.

CHOOSE ONE JOB TOGETHER

TELESCOPE SCHEDULER ☐

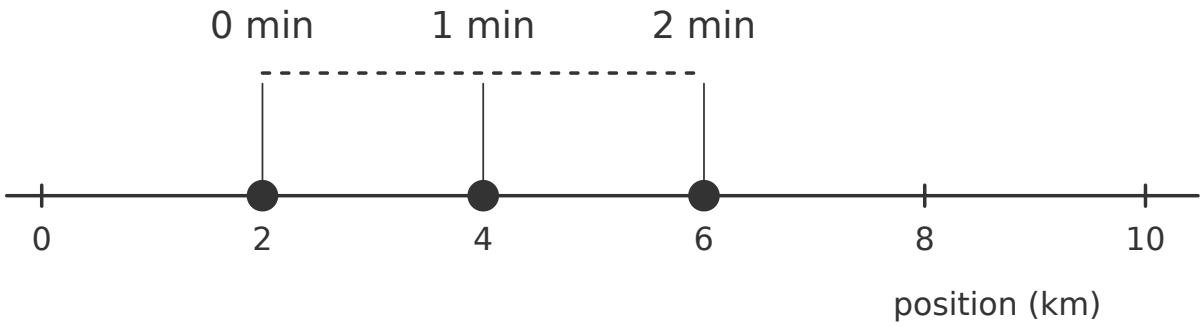
Mark where to aim at minute **3**. Tell the operator which dots were measured and which dot you predicted.

SCIENCE BROADCASTER ☐

Give viewers a two-sentence update. Point to a measured dot, then say where you expect the next one.

Word help: sighting / prediction / pattern

Three telescope fixes are at **2**, **4** and **6 km** at minutes **0**, **1** and **2**. The night crew has one more exposure.



01 OUR FIRST CALL

Where should the camera look next? Use the timed dots in your call.

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new sighting fit your prediction? Tell the next crew what to check.

Now we would say ___ because ___.

Switch speakers. What new observation would make you change your prediction? Explain why.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks